PROCESS MINING & AI

Celonis

Understanding business processes from data: a demo on a real event log

Course: Artificial Intelligence and Business Intelligence

Agenda

1 Celonis and AI

2 Our case and method in Celonis

3 Results on a real event log

4 Takeaways and limits

What is Celonis?

A platform that reads the traces left by a company's software to rebuild how its processes really work.

2011

Spin-off of the Technical University of Munich, valued at around $11 billion

Leader 2026

Gartner Magic Quadrant for Process Intelligence, the category that replaces process mining

Digital twin

The "Process Intelligence Graph" models a company's operations end to end

Sources: Celonis press release (May 2026); IT Brief; ITWire

Why Celonis talks about AI

No AI without PI

AI agents need context: they must know how a process really works before improving it.

Gartner 2026

The category is renamed from “process mining” to “process intelligence”.

30 %

of failed AI projects will be rescued by process intelligence in 2026, predicts Forrester.

“To benefit from AI, you need good data that’s well-structured, and that’s where Process Intelligence and Celonis come into play.”

Julien Nauroy, Renault Group

Sources: ITWire; IT Brief; PEX Network (Forrester)

How it works

1 · EVENT LOG

The data

A table with at least 3 columns: the **Case ID** (the file), the **Activity** (the step) and the **Timestamp** (the date).

2 · DATA MODEL

The model

We import the file, then tell Celonis which column plays which role.

3 · ANALYSIS

The result

Celonis draws the real process: steps, variants, rework loops and delays.

Nothing is drawn by hand: **everything comes from the data**.

OUR CASE

Expense reimbursement requests

In our course, Artificial Intelligence and Business Intelligence, we work on business processes. We took a public “Request For Payment” log from Kaggle (originally the BPI Challenge 2020): the journey of each request, from submission to payment.

6,886	36,796	2017–19	89
cases	events	period covered	different paths

What we did in Celonis

1

Data Pool

Created a data workspace called “BPIC2020”

2

Import

Converted the .xes file to CSV, uploaded it, set Timestamp as a date

3

Data Model

Chose Case ID, Activity and Timestamp, then loaded it

4

Studio

Created an Analysis linked to the model

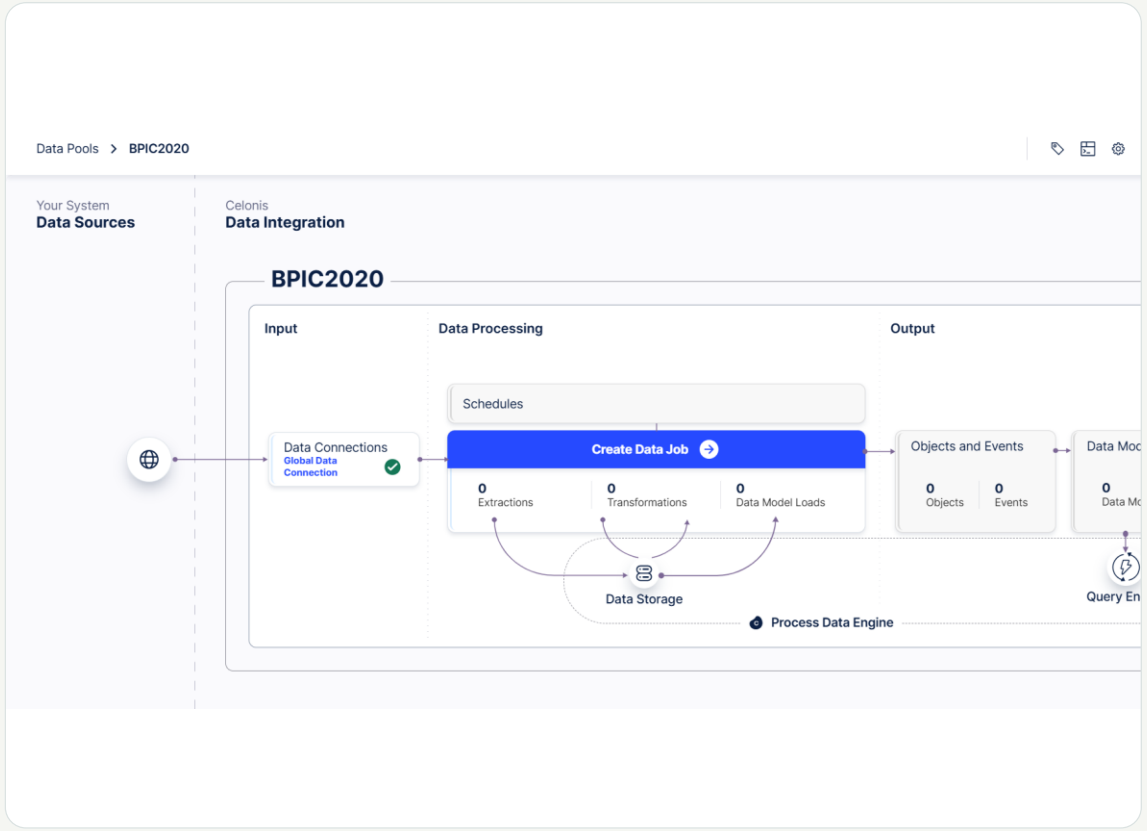
5

Process Explorer

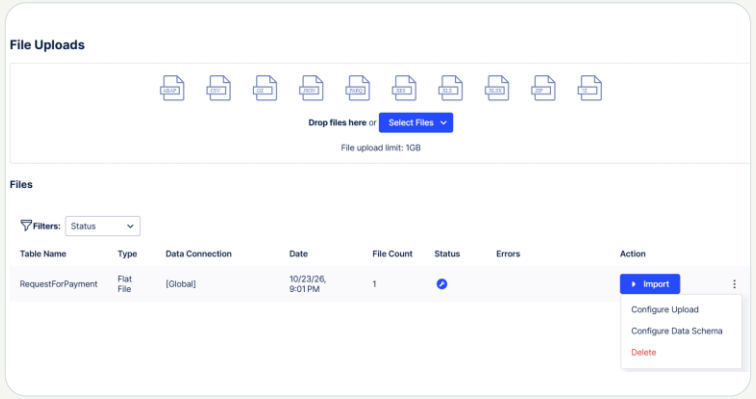
The process map appeared automatically

No diagram drawn by hand: it all comes from the data.

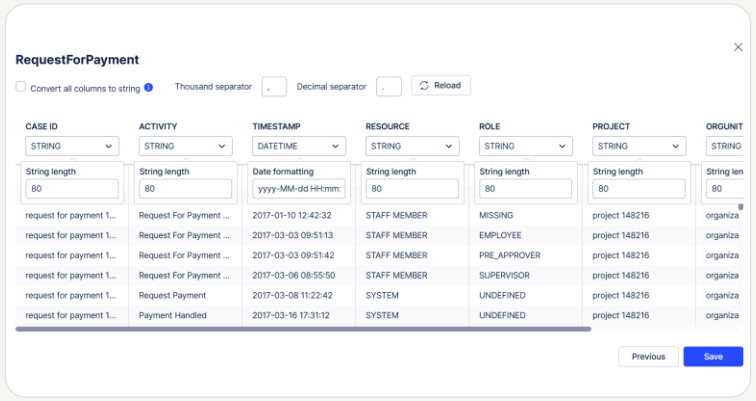
Step by step in Celonis (1/2): getting the data in



1 · Data Pool “BPIC2020” created

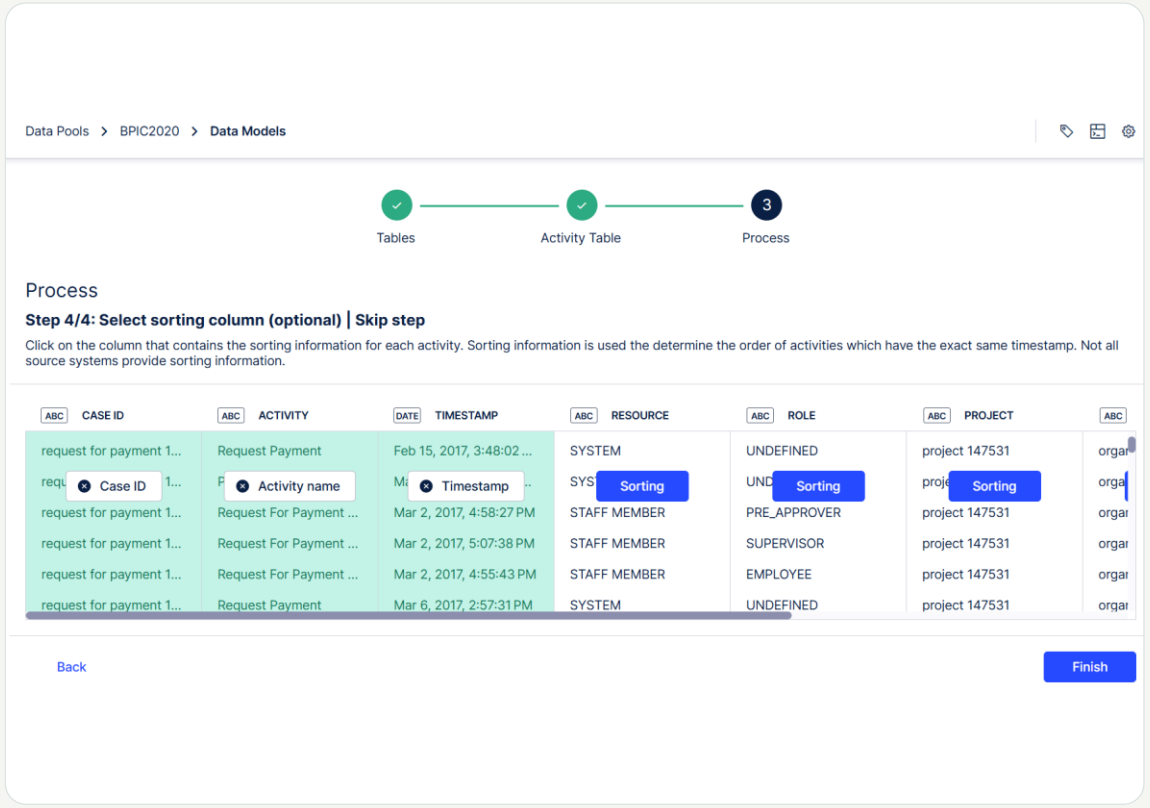


2 · CSV file uploaded

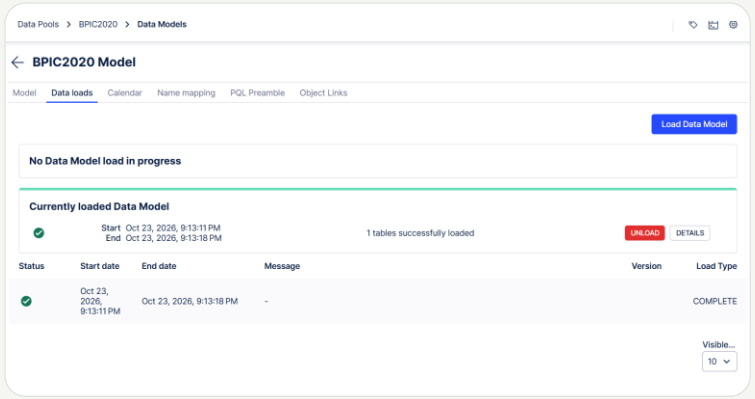


3 · Timestamp set as a date (DATETIME)

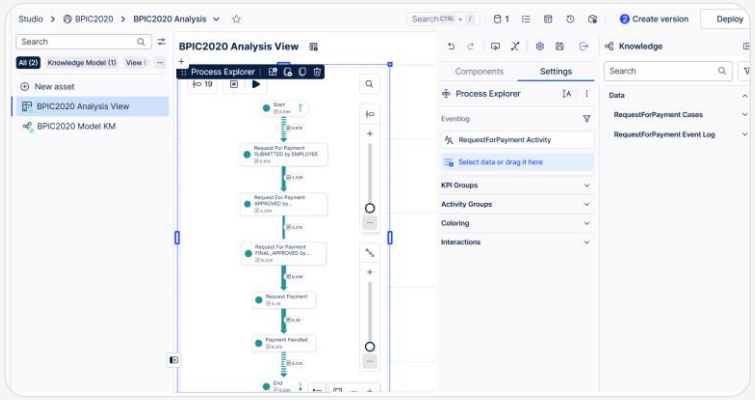
Step by step in Celonis (2/2): from model to map



1 · Case ID, Activity and Timestamp assigned in the Data Model



2 · Data Model loaded successfully



3 · Process Explorer added in Studio

The process, step by step: who does what

1 · Employee
Submits the request

2 · Administration
Checks and approves it

3 · Budget owner
Validates it too, in some cases (about 2,000)

4 · Supervisor
Gives the final approval

5 · System
Requests the payment

6 · System
Makes the payment

Exceptions: a rejection sends the request back to the employee, and some cases skip a step.

The process at a glance

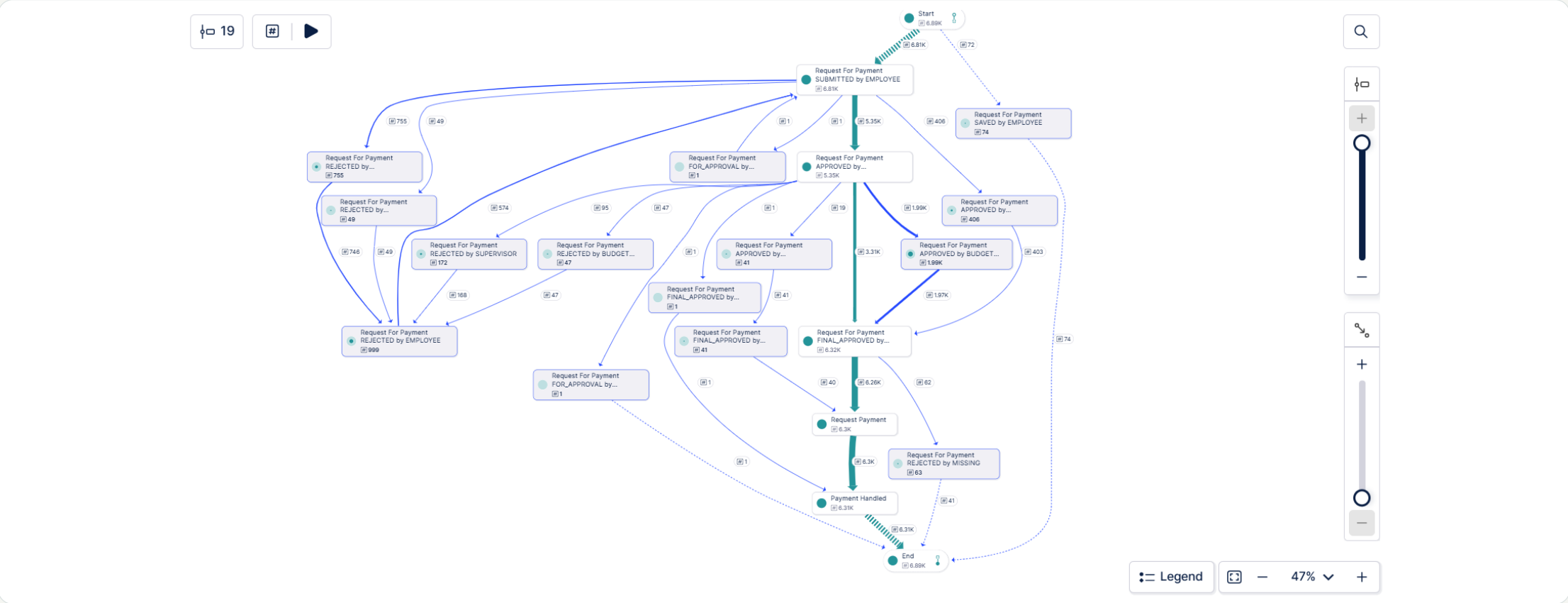
Each box is a step, each arrow a transition. The thickness of the arrow shows how many cases go through it.

- The employee submits the request
- Successive approvals
- Payment is requested, then made

About **6,300 payments** for 6,900 incoming cases.



The real process, with its detours

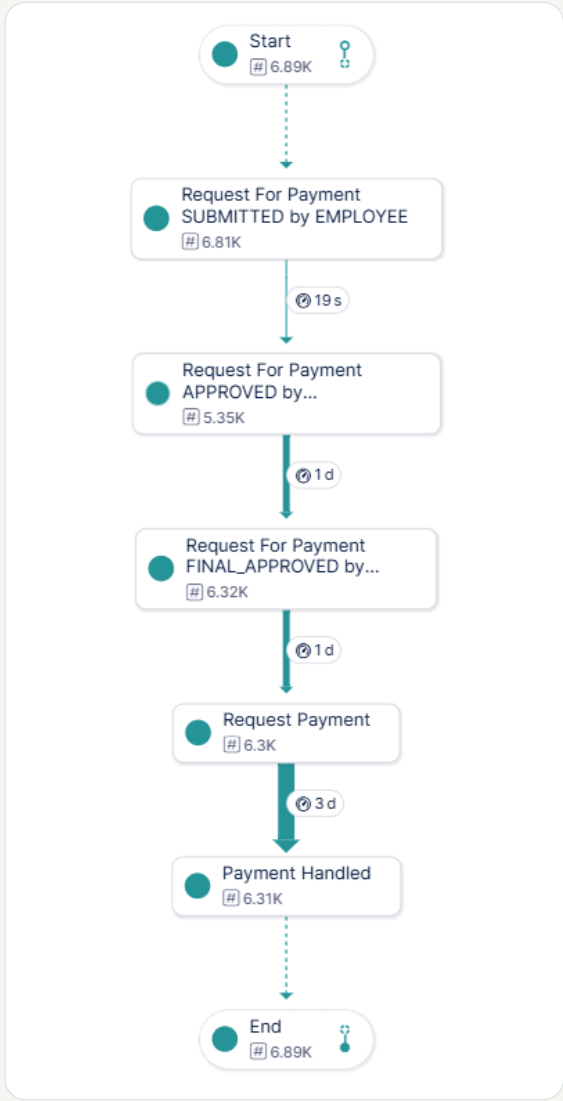


With more activities, the exceptions appear: rejection loops, extra validations, saved drafts.

How long does each step take?

19 sec	Submission → approval
1 day	Approval → final approval
1 day	Final approval → payment request
3 days	Payment request → payment made

Median time between steps, as shown by Celonis. On the main path, the longest wait is at the end.

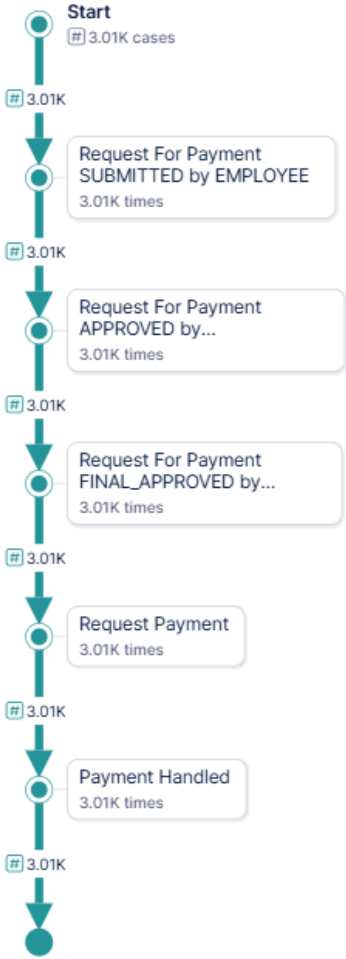


The most frequent path

Of the 89 variants, this one is the most common: no rejection, no detour.

44 % of the cases: 3,010 out of 6,886

The other 56 % follow a longer path: budget validation, rejections, resubmissions.



What Celonis reveals

≈ 1,000

Rejections by the employee

Requests to correct and resend: extra steps and rework.

≈ 2,000

Budget validations

An extra step that only part of the cases go through.

These gaps are what Celonis spots, so that people can investigate why they happen.

Takeaways

See what is real	Celonis shows how processes really run, not how they should run.
Feed the AI	The process map gives context to a company's AI agents.
Limits	Everything depends on data quality. Our demo used a public Kaggle dataset, not real company data.

Thank you. Questions?